AI for Cyber Security Group Project

1. Problem, Dataset, Pair-Level Labelling

This project builds a machine-learning workflow for spotting likely online grooming pairs in the PASYDA synthetic metadata dataset. The data does not include message content. It gives only communication metadata: ID, Source, Destination, Date, and Length.

The main methodological choice was to treat each central-node/contact-node pair as the supervised sample. This matters because the solution files name the true pair for each scenario. They do not label each message row.

The original PASYDA files are not directly labelled for supervised learning. Labels were created by matching each scenario's solution file to the generated central-node/contact-node candidate pairs. This produced one positive pair and ten negative pairs per scenario.

Table 1: PASYDA Dataset and Pair-Level Modelling Summary

Dataset folders	5 folders, Dataset-1 to Dataset-5
Scenarios	10 scenarios in total, with 2 scenarios in each folder
Candidate pairs	11 candidate central-node/contact-node pairs in each scenario
Positive labels	1 true grooming-related pair in each scenario
Final modelling rows	110 pair-level samples overall: 10 positive and 100 negative
Evaluation grouping	One complete dataset folder held out in each fold

Training on raw message rows would have been the wrong setup, because the labels belong to pairs. We therefore used the raw CSV files to build behavioural features for each candidate pair, then trained the models on the pair-level table.

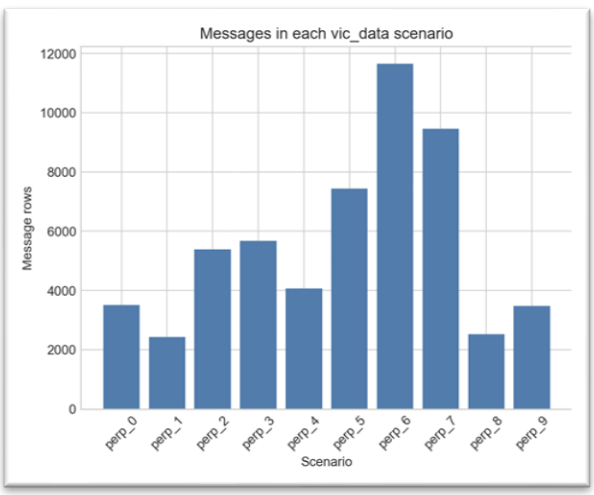

Fig. 1: Message Volume Across PASYDA Scenarios

Fig. 2: Pair-Level Class Imbalance in the Modelling Dataset

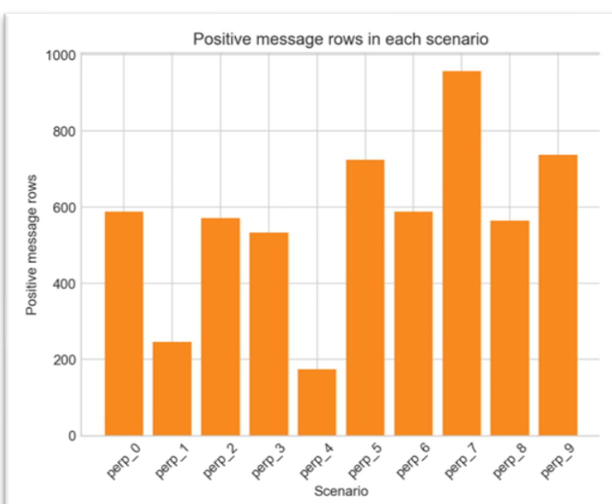

Fig. 3: Positive Message Rows Across Scenarios

2. Feature Engineering and Leakage-Safe Evaluation

The feature engineering step turned the original metadata into behaviour-based measures for each candidate pair. The frozen base table keeps the core interaction statistics. The enhanced version adds scenario context and simple sequence-based features.

Table 2: Engineered Feature Groups and Their Purpose

Feature group	Examples	Purpose
Volume and balance	message_count, sent_ratio, reciprocity_ratio	Measure how much the pair communicated and who drove the exchange
Timing	duration_hours, evening_ratio, late_night_ratio	Capture when the interaction happened and how long it lasted
Message length	mean_length, short_message_ratio	Check whether shorter messages carry useful signal
Scenario context	within-scenario ranks and z-scores	Compare each pair with the other candidates in the same scenario
Sequence behaviour	first_day_intensity, last_day_intensity, direction_switch_frequency	Describe how the exchange changed over time

Evaluation used grouped 5-fold testing by dataset folder. In each fold, the model trained on four full folders and tested on the remaining folder, so every split had 88 training rows and 22 test rows.

The pair-level dataset is heavily imbalanced, with 10 positives and 100 negatives. Random Forest (RF) and Support Vector Machine (SVM) used balanced class weights. The Multilayer Perceptron (MLP) used positive-class oversampling, but only inside the training folds. The test folds were left untouched. For models that are sensitive to feature scale, such as SVM and MLP, numerical features were scaled using training-fold statistics only, so information from the held-out folder was not used during preprocessing.

Table 3: Evaluation Metrics Used for Model Comparison

Metric	Why it is used
Top-1 Accuracy	Shows whether the true pair was placed first in each scenario
Mean Reciprocal Rank (MRR)	Gives credit when the true pair is near the top, even if it is not ranked first
Precision / Recall / F1 at Top-1	Measures false alerts and missed detections for the selected top pair
ROC-AUC and PR-AUC	Evaluate ranking quality when positives are rare

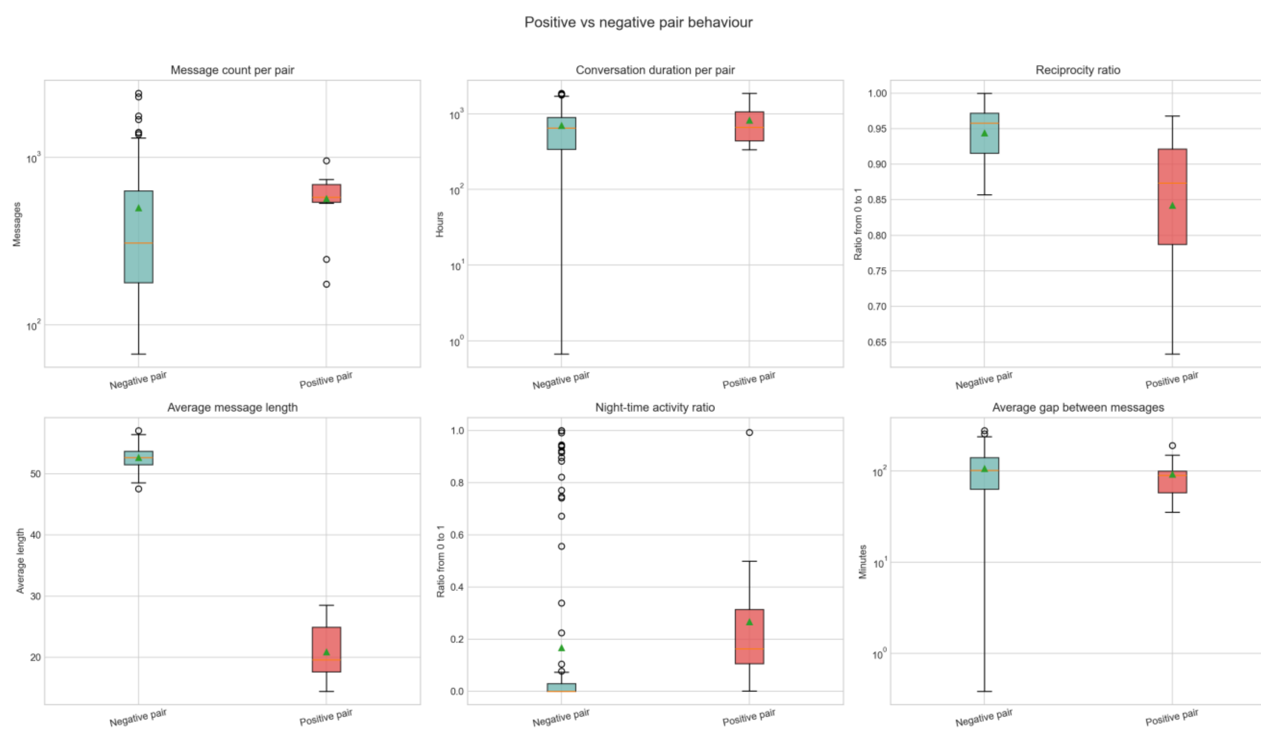

Fig. 4: Behavioural Differences Between Positive and Negative Pairs

3. Experiment Portfolio and Model Comparison

The experiment portfolio contained 11 separate experiment reports: two heuristic baselines, three RF experiments, three SVM experiments, and three MLP experiments. All experiments used the same pair-level dataset, grouped folder split, and evaluation metrics, so the results could be compared fairly.

Three model families were trained under the same grouped evaluation protocol: RF, RBF-kernel SVM, and a small MLP. We used an MLP because the final data is tabular and the labelled dataset is too small to justify sequence models such as LSTMs or Transformers.

RF was used because it works well on small tabular datasets and can model non-linear feature interactions. SVM was included as a strong classical margin-based model for small datasets. The MLP was used as the deep learning comparison, but it was kept lightweight because the final representation is tabular rather than sequential text.

The most important result is also the easiest one to miss: choosing the pair with the shortest average message reached the same perfect Top-1 score as the full-feature trained models. That does not make the models useless. It means PASYDA has a very strong synthetic signal, so the perfect scores need to be read carefully.

Hyperparameters were checked with compact nested grouped validation inside each outer training split. The tested RF depth and leaf-size settings, SVM C values, and MLP hidden sizes and regularisation values all kept perfect full-feature test performance. For that reason, the interpretation focuses on robustness rather than trying to squeeze out a higher score. The held-out dataset folder was not used for tuning, oversampling, feature selection, or threshold selection. This reduced leakage risk and kept the model comparison fair.

Table 4: Results Across the 11 Experiment Reports

Experiment	Top-1	MRR	F1	ROC-AUC	PR-AUC
EXP-01 Highest message count baseline	0.00	0.31	0.00	0.68	0.26
EXP-02 Lowest mean length baseline	1.00	1.00	1.00	1.00	1.00
EXP-03 Random Forest base	1.00	1.00	1.00	1.00	1.00
EXP-04 Random Forest contextual	1.00	1.00	1.00	1.00	1.00
EXP-05 Random Forest no length	0.70	0.80	0.70	0.96	0.84
EXP-06 SVM base	1.00	1.00	1.00	1.00	1.00
EXP-07 SVM contextual	1.00	1.00	1.00	1.00	1.00
EXP-08 SVM no length	1.00	1.00	1.00	1.00	1.00
EXP-09 MLP base	1.00	1.00	1.00	1.00	1.00
EXP-10 MLP contextual	1.00	1.00	1.00	1.00	1.00
EXP-11 MLP no length	0.90	0.91	0.90	0.93	0.91

The **11 experiments** show that the full-feature models perform perfectly on PASYDA, but the baselines and no-length ablations make the interpretation more cautious. The lowest-mean-length baseline also reached perfect performance, which suggests that message length is a strong synthetic cue in this dataset. The no-length results are therefore important because they test whether the models still work when that cue is removed. Precision and recall were calculated at Top-1, and the results table reports F1 as their combined summary measure.

Each experiment was recorded as a separate report, with the same dataset version, fold structure, and metric definitions used throughout the portfolio.

RF, SVM, and MLP all reached perfect scores with the base and contextual feature sets. The no-length ablation is more revealing: RF and MLP lost some performance when length cues were removed, while SVM stayed stable.

The baselines add useful context. Highest message count scored 0.00 Top-1 and longest duration scored 0.30, while lowest mean length and highest short-message ratio both scored 1.00.

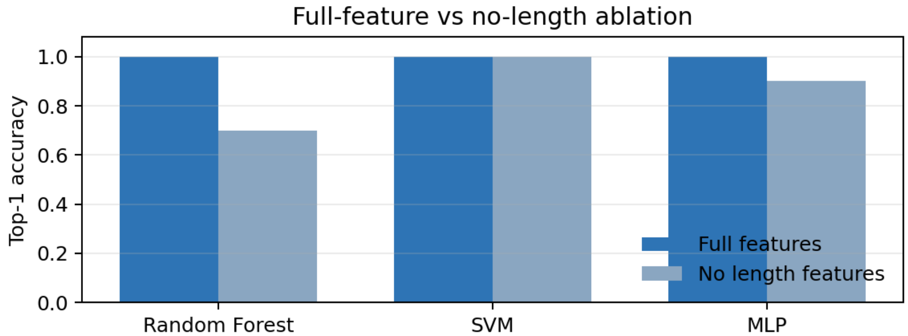

Fig. 5: Full-Feature and No-Length Ablation Performance Across Models

4. Critical Analysis, Limitations, and Future Work

The results are strong on PASYDA, but they should not be treated as evidence that grooming detection is solved in real settings. The dataset is synthetic, small, and contains only 10 positive pair-level samples, so the reported scores need careful interpretation.

The no-length runs produced concrete errors. Random Forest missed 3 of the 10 true pairs, and the MLP missed 1. The shared difficult case was perp_4 in Dataset-3, where both RF and MLP failed once length-derived cues were removed. These errors suggest that some true grooming pairs are only easy to identify when message-length cues are available. Without those cues, the models need stronger behavioural or contextual metadata to separate the true pair from similar candidates.

Limitations

- The main limitation is that the dataset appears to contain a strong synthetic pattern.
- The message length baselines performed perfectly, which suggests that some results may be linked to artefacts in the generated data rather than behaviour that would always appear in real cases. To address this, the no-length ablation should be reported clearly, and the approach should be tested on external data before making stronger claims.
- False positives are important because a wrongly flagged pair could waste investigator time and may unfairly affect innocent users.
- False negatives also carry serious risk because a genuinely harmful interaction could be missed if the model fails to rank the true pair highly enough. This is why the evaluation should not rely on accuracy alone. Ranking measures such as MRR, along with recall-focused analysis, are more useful for understanding whether harmful pairs are being pushed near the top.
- The dataset also has limited metadata. It does not include message content, account age, platform context, user history, or reporting outcomes. This makes the task less realistic and limits what the models can learn.

Future Work

Future work should collect richer features while still protecting privacy. Account age could help identify newly created profiles. Platform context could separate private direct messages from public or group activity. Report status and moderation outcomes could provide stronger weak labels, and longitudinal annotations could show whether behaviour escalates over time.

A stronger evaluation would also need larger labelled datasets, more positive examples, and repeated validation across independent data. The positive class is currently very small, with only 10 positive pair-level samples, so the reported scores should be treated with caution.

In practice, this system should be used as a ranking and triage tool rather than a final decision-maker. The proposed pipeline avoids the main leakage risk and performs strongly on PASYDA, but any real-world use should be framed as decision support for human review, not as an automated accusation system.